Join

- **Join** is a combination of a Cartesian product followed by a selection process. A Join operation pairs two tuples from different relations, if and only if a given join condition is satisfied.
- We will briefly describe various join types in the following sections

Theta (θ) Join

- Theta join combines tuples from different relations provided they satisfy the theta condition. The join condition is denoted by the symbol **θ**.
- **The θ-Join of two relations *R* and *S* is denoted as:**
  - $R \bowtie_F S$
  - **Where *F* is the formula, specifying the join predicate.**
  - **A join predicate specified, is similar to selection predicate except the terms are of the form "R.AθS.B" where A and B are attributes of 'R' and 'S' respectively.**
  - **The join of two relations is equivalent to performing a selection, using the join predicate as the selection formula, over the Cartesian product of the two operand relations.**
  - $R \bowtie_F S = \sigma_F (R \times S)$

θ-Join -Example

- **EMP**

ENO	ENAME	TITLE
E1	J. Doe	Elect Eng
E2	M. Smith	Syst. Anal.
E3	A. Lee	Mech. Eng

PAY

TITLE	SALARY
Elect Eng	50000
Syst. Anal.	55000
Programmer.	65000

$\textbf{EMP} \bowtie_{\textbf{EMP.TITLE=PAY.TITLE}} \textbf{PAY}$

ENO	ENAME	TITLE	SALARY
E1	J. Doe	Elect Eng	50000
E2	M. Smith	Syst. Anal.	55000

θ-Join -Example

- **If the attributes of two relations are common then projection is necessary to make sure that those attributed do not appear twice in the result.**
- **So we can write above as:**

$$\text{EMP} \bowtie_{\text{EMP.TITLE=PAY.TITLE}} \text{PAY} =$$

$$\Pi_{\text{ENo, ENAME, TITLE, SAL}}(\sigma_{\text{EMP.TITLE=PAY.TITLE}}\ (\text{EMP x PAY})$$

- **This example demonstrates a special case of θ-join which is called equi-join. Because the join predicate contains equality (=) as arithmetic operator.**

Equijoin

- When Theta join uses only **equality** comparison operator, it is said to be equijoin. The above example corresponds to equijoin.

Natural Join (⋈)

- Natural join does not use any comparison operator. It does not concatenate the way a Cartesian product does. We can perform a Natural Join only if there is at least one common attribute that exists between two relations. In addition, the attributes must have the same name and domain.
- Natural join acts on those matching attributes where the values of attributes in both the relations are same.

Example

Courses		
CID	Course	Dept
CS01	Database	CS
ME01	Mechanics	ME
EE01	Electronics	EE

HoD	
Dept	Head
CS	Alex
ME	Maya
EE	Mira

Courses ⋈ HoD			
Dept	CID	Course	Head
CS	CS01	Database	Alex
ME	ME01	Mechanics	Maya
EE	EE01	Electronics	Mira

INNER JOIN

- INNER JOIN selects records that have matching values in both tables as long as the condition is satisfied. It returns the combination of all rows from both the tables where the condition satisfies.

Example

Employee				
EMP_ID	EMP_NAME	CITY	SALARY	AGE
1	Angelina	Chicago	200000	30
2	Robert	Austin	300000	26
3	Christian	Denver	100000	42
4	Kristen	Washington	500000	29
5	Russell	Los angels	200000	36
6	Marry	Canada	600000	48

Project		
PROJECT_NO	EMP_ID	DEPARTMENT
101	1	Testing
102	2	Development
103	3	Designing
104	4	Development

Output

EMP_NAME	DEPARTMENT
Angelina	Testing
Robert	Development
Christian	Designing
Kristen	Development

Outer Joins

- Theta Join, Equijoin, and Natural Join are called inner joins. An inner join includes only those tuples with matching attributes and the rest are discarded in the resulting relation. Therefore, we need to use outer joins to include all the tuples from the participating relations in the resulting relation. There are three kinds of outer joins − left outer join, right outer join, and full outer join.

Left Outer Join($R ⟕ S$)

- All the tuples from the Left relation, R, are included in the resulting relation. If there are tuples in R without any matching tuple in the Right relation S, then the S-attributes of the resulting relation are made NULL.

Example

Left Table	
A	B
100	Database
101	Mechanics
102	Electronics

Right Table	
A	B
100	Alex
102	Maya
104	Mira

Output			
A	B	C	D
100	Database	100	Alex
101	Mechanics	---	---
102	Electronics	102	Maya

Right Outer Join(R⟖ S)

- All the tuples from the Right relation, S, are included in the resulting relation. If there are tuples in S without any matching tuple in R, then the R-attributes of resulting relation are made NULL.

Example

Left Table	
A	B
100	Database
101	Mechanics
102	Electronics

Right Table	
A	B
100	Alex
102	Maya
104	Mira

Output			
100	Database	100	Alex
102	Electronics	102	Maya
---	---	104	Mira
100	Database	100	Alex

Full Outer Join (R ⟗ S)

- All the tuples from both participating relations are included in the resulting relation. If there are no matching tuples for both relations, their respective unmatched attributes are made NULL.

Example

Left Table	
A	B
100	Database
101	Mechanics
102	Electronics

Right Table	
A	B
100	Alex
102	Maya
104	Mira

Output			
A	B	C	D
100	Database	100	Alex
101	Mechanics	---	---
102	Electronics	102	Maya

Task No. 3

Submission Date: Next Class

Note: Assignments must be hand written. Examples should not be copied from each other.

Q.No. 1 : Explain Following Relational Algebra operations with suitable examples.

1. Natural Join
2. Semi Join
3. Theta Join